Pal 2021 breast cancer — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

iLISI 1.00→4.95 of 22 samples (per-neighbourhood, θ=10); one connected component on the independent Harmony-PCA and diffusion-map graphs. Mixed and continuous.

Gate 2 — module geography → GO (non-branching)

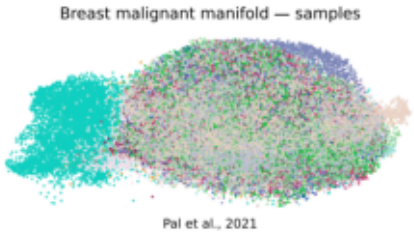
Single organized axis: pSom/Gast-high origin → Crest-high terminus (Crest decile pt 0.93; pSom&Crest enrich 0.23, inverse terminal gradient). Non-branching — no Gast/pSom co-expression hub (enrich 1.03).

Gate 3 — cross-repression → NA

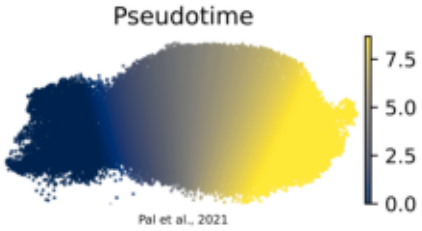
No bifurcation (no hub, no second arm) — not applicable. Gast & pSom show mild spatial cross-repression (smoothed r=−0.32 on the Body), permitted in the non-branching case.

STEP FIGURES

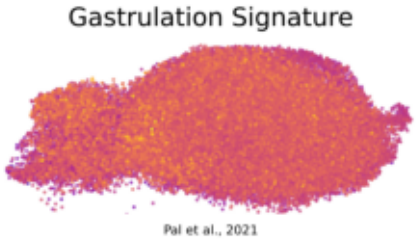
Manifold — samples



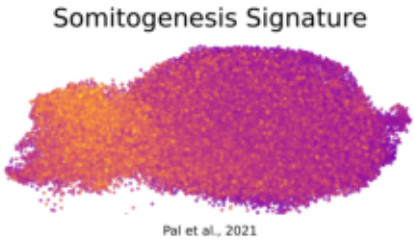
Pseudotime



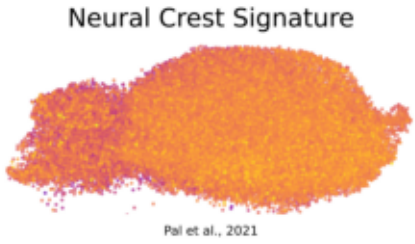
Gastrulation



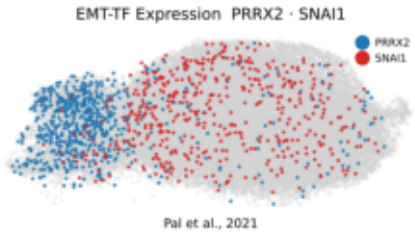
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast cancer (Pal et al., 2021), 10x — 59,367 malignant cells (22/26 samples); trajectory scoped to the 44,398-cell low-cycling Body. Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form — one continuous pSom/Gast → Crest developmental axis, no Gast/pSom bifurcation. G1 GO / G2 GO / G3 NA. EMT-TFs PRRX2 (origin) · SNAI1 (latest) — all 5 detected factors peak at the origin, none Crest-terminal.

Parameter choices

Min-cell threshold	12 (geom-mean of 56→3 cliff); keeps 22/26 samples	**Integration**	Harmony θ=10, n_pcs=20 (iLISI plateau 5–10)
Geometry	Linear (non-branching); scFates curve N=20, 2 tips / 0 forks	**Root**	pSom/Gast-high, Crest-low origin; terminus Crest-high
Scope (cell-cycle)	Body 44,398 (low-cycling); 14,965 proliferative compartment excluded	**Module means (all)**	Gast −0.007 · pSom +0.036 · Crest −0.034; hub enrich 1.03 (none)
Cycle independence	pt vs proliferation ρ=0.10 (cycle-independent axis)	**EMT-TFs**	PRRX2 (origin) · SNAI1 (latest); all 5 detected peak at origin